

Individual differences in dynamic belief updating during trust learning



PROGRAM IN
NEUROSCIENCE &
COGNITIVE SCIENCE



V. Guigon^{1,2}, S. Topel³ and Caroline J. Charpentier^{1,2,4}

1. Department of Psychology, University of Maryland, College Park, MD, USA
2. Program in Neuroscience And Cognitive Science, University of Maryland, College Park, MD, USA
3. Faculty of Social and Behavioral Sciences, Leiden, Netherlands
4. Brain and Behavior Institute, University of Maryland, College Park, MD, USA



Universiteit
Leiden
Social and Behavioural Sciences



BRAIN & BEHAVIOR
INSTITUTE

CONTEXT

- **Trust** is fundamental to social interaction. It is defined as the **willingness to be vulnerable to another being on the basis of positive expectations** of their intentions and behaviors (1).
- In repeated interactions, **learning to trust others** involves **cognitive processes** that integrate **uncertainty, context, and potential betrayal**.
- **Different strategies** can guide trust behavior. Some rely on **heuristics** (e.g., fixed rules or triggers). Others use **reinforcement learning**, adjusting expected value through associative updates, or **Bayesian belief updating**, which integrates uncertainty into probabilistic inferences about partners' intentions (2, 3).
- **Individuals differ** in how they deploy cognitive **processes**, which may impact their **strategies**. Yet, this heterogeneity has not been systematically characterized. Here, we aim to do so:
 - How does dynamic trust learning integrate sensitivity to betrayal, uncertainty and context?
 - How do these processes vary across individuals and reveal distinct cognitive profiles?

HYPOTHESES

• Gain/loss learning asymmetry

Higher learning rate after losses than gains.

• Individual differences

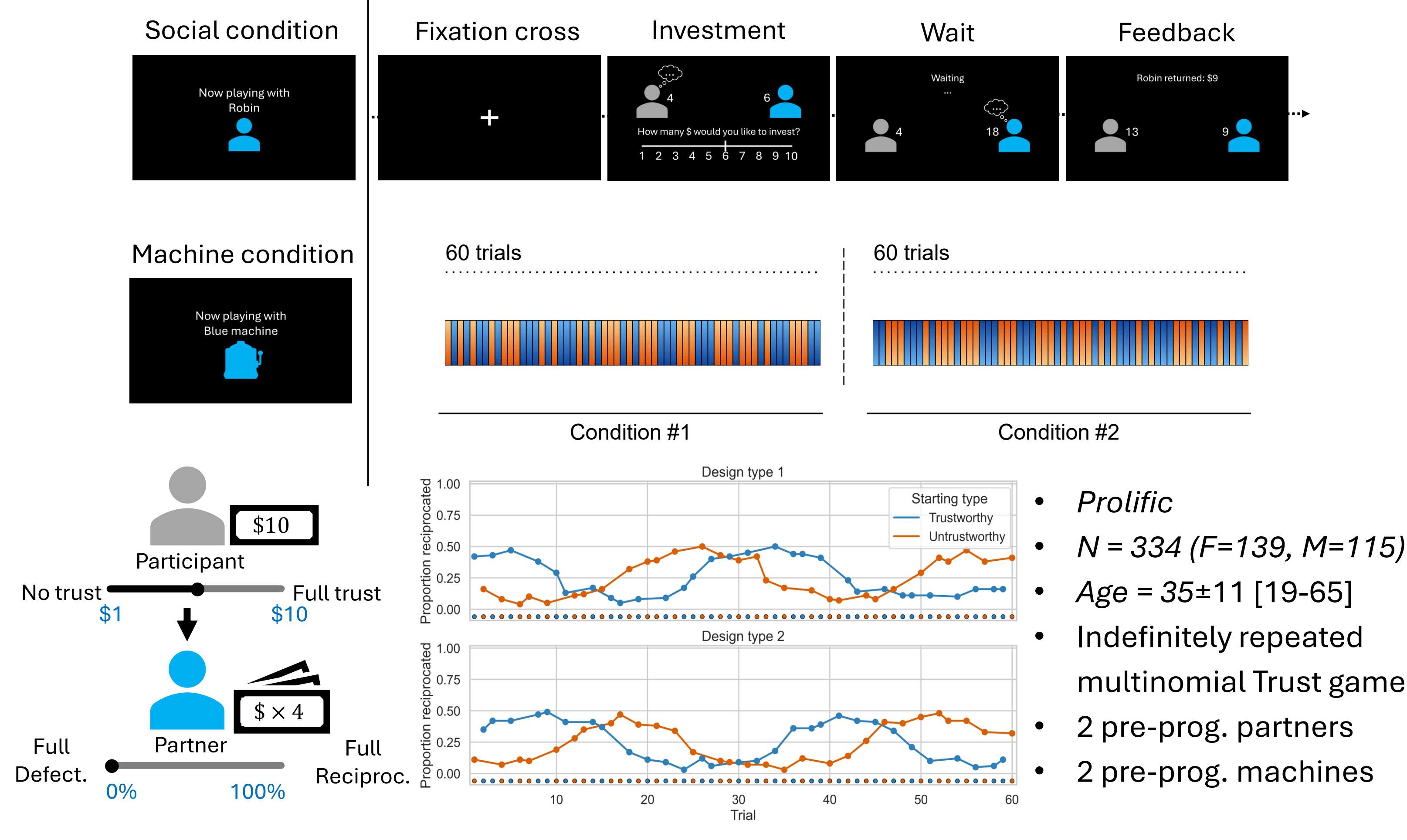
- A subset of participants deploy **heuristics** (non-learners).

• Context modulation

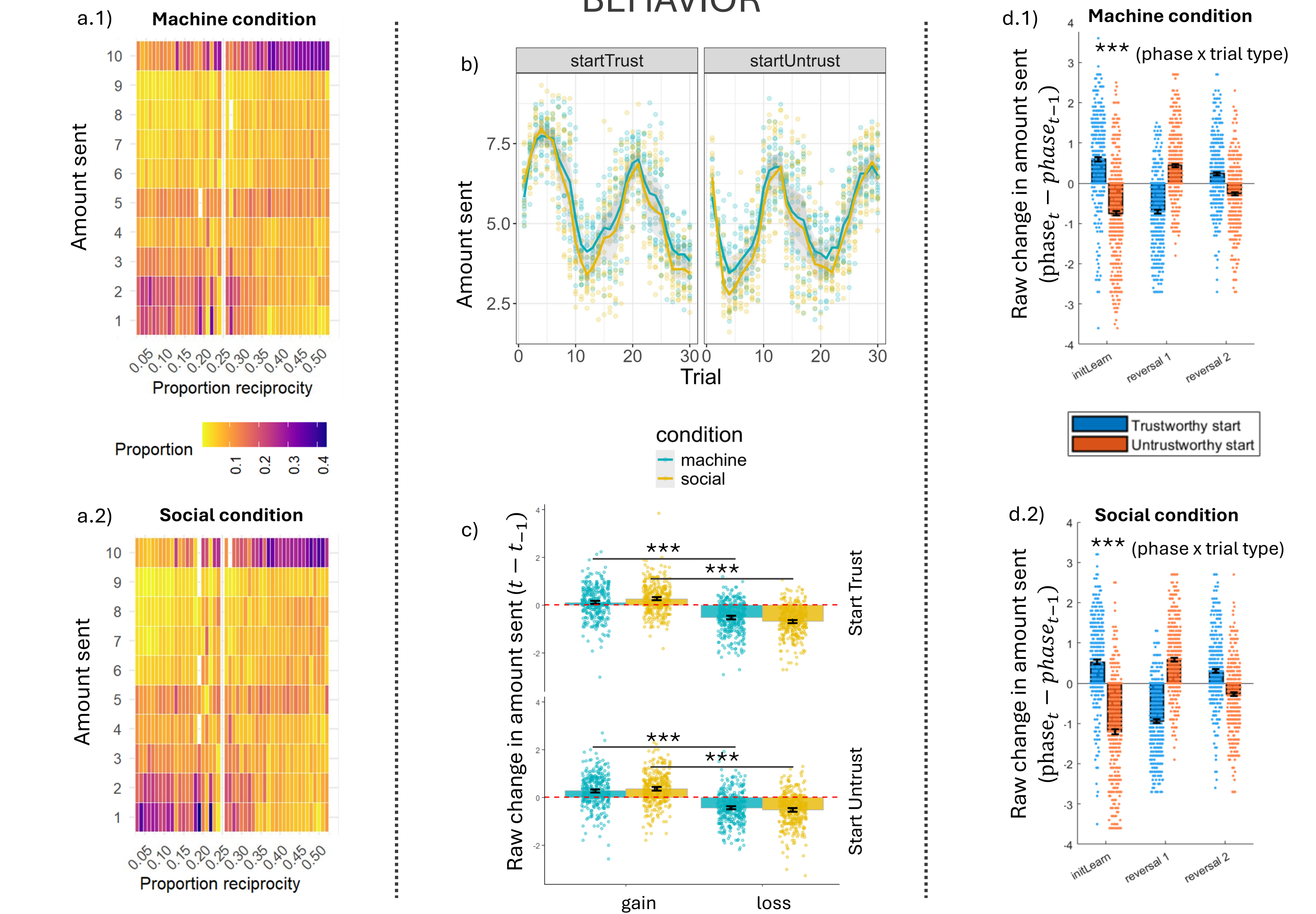
Betrayal aversion leads to **stronger learning rates after losses in the social condition**.

- Among learners, **variability** in learning asymmetry and betrayal aversion is captured by model fit/parameters

METHODS



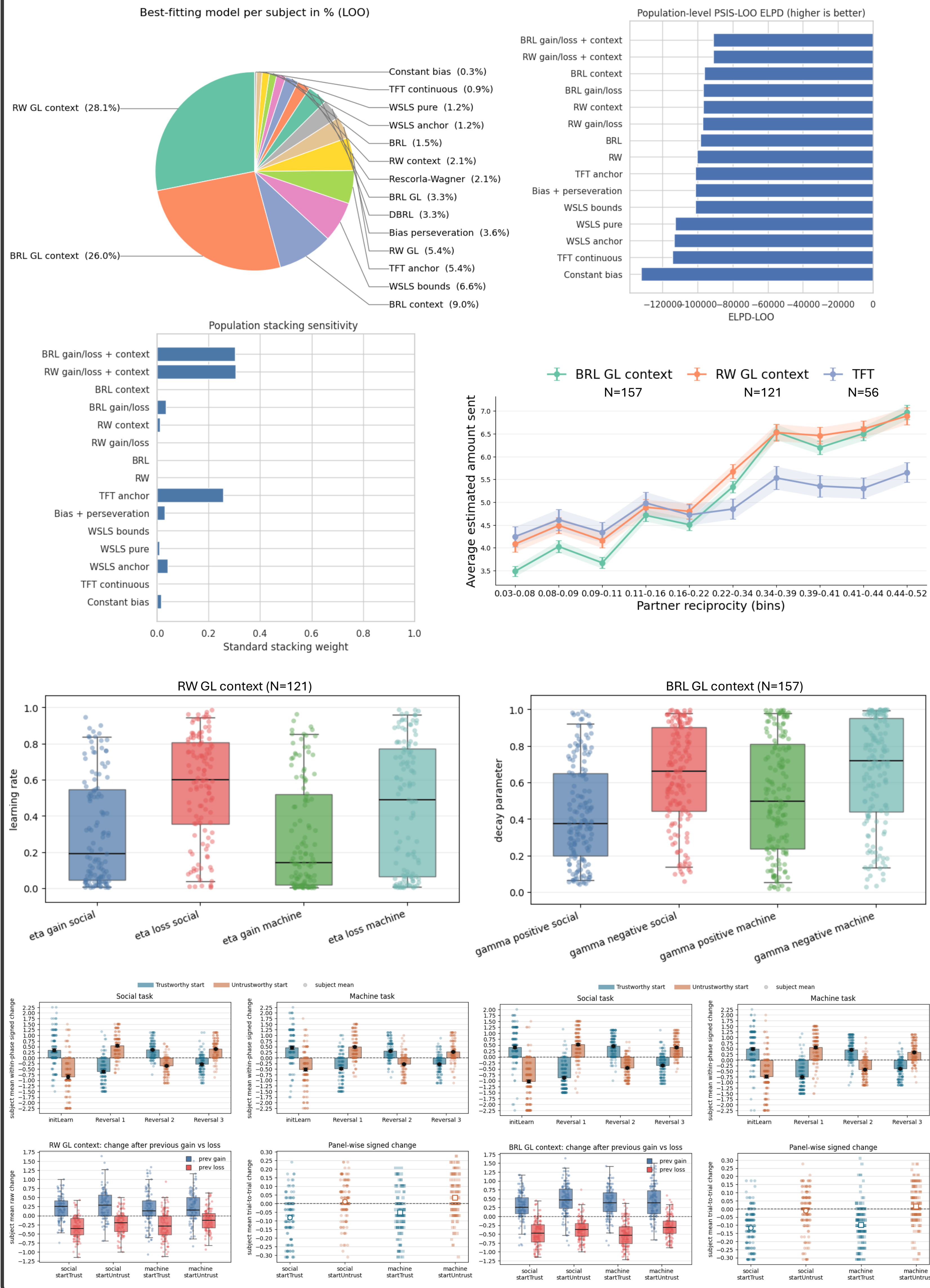
BEHAVIOR



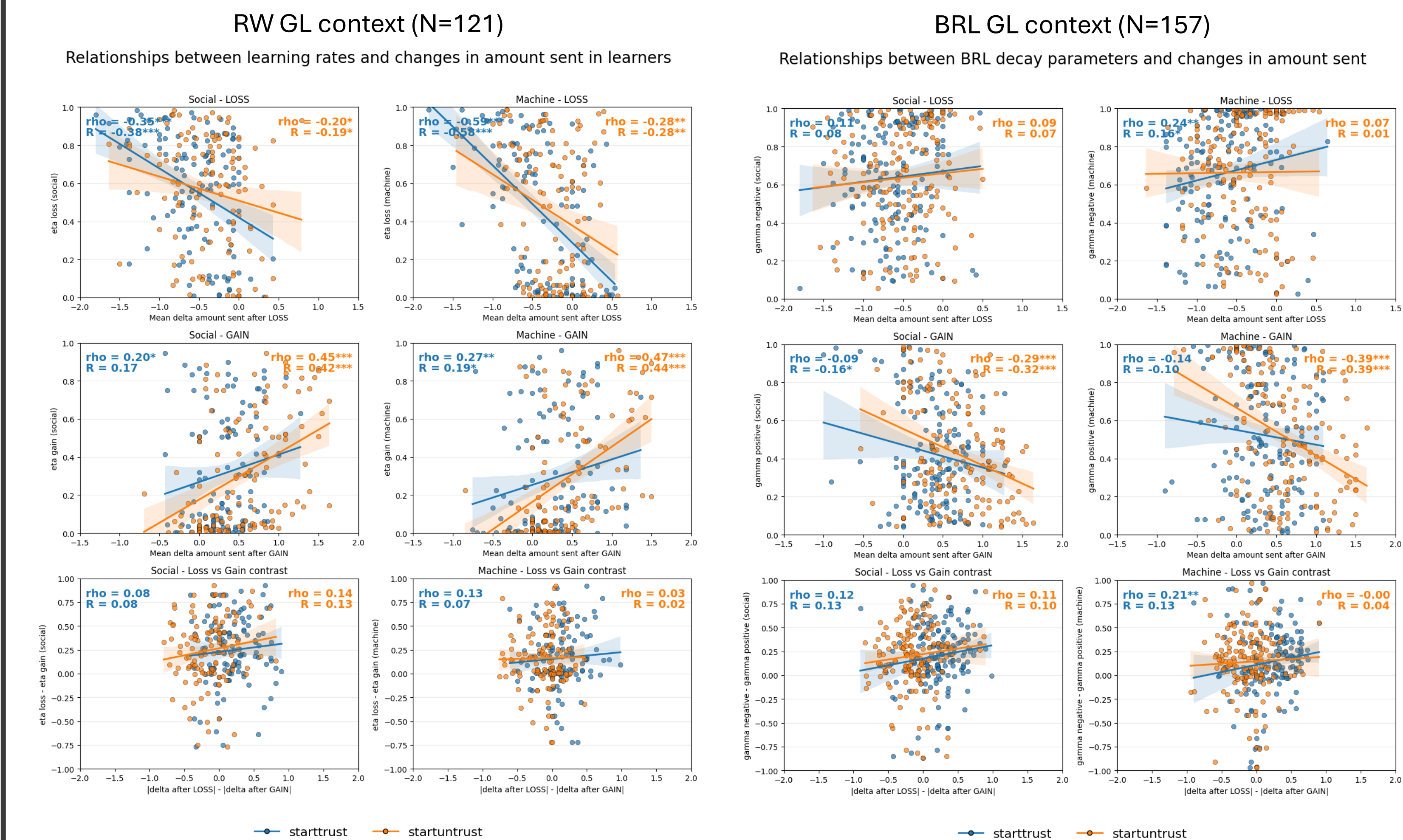
CONCLUSIONS & NEXT STEPS

- Trust behavior shows **gain/loss asymmetry, context modulation, and individual variability** in strategy use.
- Best-fitting model was **BRL GL context**, but sample is better explained by a roster of 3 models, including a heuristic.
- Among learners, the models captures different variabilities in betrayal aversion & context sensitivity. **Loss** (and gain) learning parameters correlated differently with changes in amount sent after losses (and gains) in both conditions.
- Next, we are extending the model space to include **Bayesian updating with sensitivity to uncertainty** (4).
- We will examine links between model parameters and individual-difference measures, improve the allocation method, and validate findings in a replication dataset.

GAIN-LOSS AND CONTEXT DEPENDENCY



INDIVIDUAL DIFFERENCES



REFERENCES

- (1) Alós-Ferrer, C., & Farolfi, F. (2019). *Frontiers in neuroscience*, 13, 887.
- (2) Bellucci, G., & Dreher, J. C. (2021). *The Neurobiology of Trust*, Cambridge University Press, Cambridge, 185-218.
- (3) Engle-Warnick, J., & Slonim, R. L. (2006). *Economic theory*, 28(3), 603-632.
- (4) Lamba, A., Frank, M. J., & FeldmanHall, O. (2020). *Psychological science*, 31(5), 592-603.

- Forming trust in others is transversal to many social situations, and **updating** trust is critical when **interactions are repeated**. Trust can be defined as the **willingness to be vulnerable** to another on the basis of **positive expectations** of the other's intentions and behaviors (Alós-Ferrer & Farolfi, 2019).

- Decisions are **risky** when made in environments with uncertainty caused by nature. They are **trusting** when outcomes depends on uncertain decisions of others. The source of uncertainty matters as individuals are **averse to betrayal** (Aimone, Houser & Weber, 2014).

- Repeated, dynamic trust situations involve strategic interactions, hidden intentions, and noisy information. Formally, the trust game can be cast as a **Bayesian game**, where players should update beliefs about their partners' intentions. Optimal responses require stochasticity, as deterministic play is easily exploited.

- At the same time, there is no single “best” way to act. Multiple strategies can yield good outcomes depending on prior expectations, dispositions, and social preferences. Empirical evidence suggests that people rely on a limited repertoire that includes **rule-based trigger strategies** as well as **learning- or mentalizing-based approaches** (Engle-Warnick & Slonim, 2006; Bellucci & Dreher, 2021). Learning to trust therefore draws on both social cognitive skills and domain-general learning skills.

- Individuals vary in their ability to deploy these processes, yet systematic **characterization of such variability remains limited**. People differ in their tolerance for uncertainty, social learning skills, mentalizing capacities, and sensitivity to betrayal, and in the biases shaping their beliefs. Such heterogeneity make trust a powerful domain for studying how multiple cognitive mechanisms jointly determine adaptive social behavior.

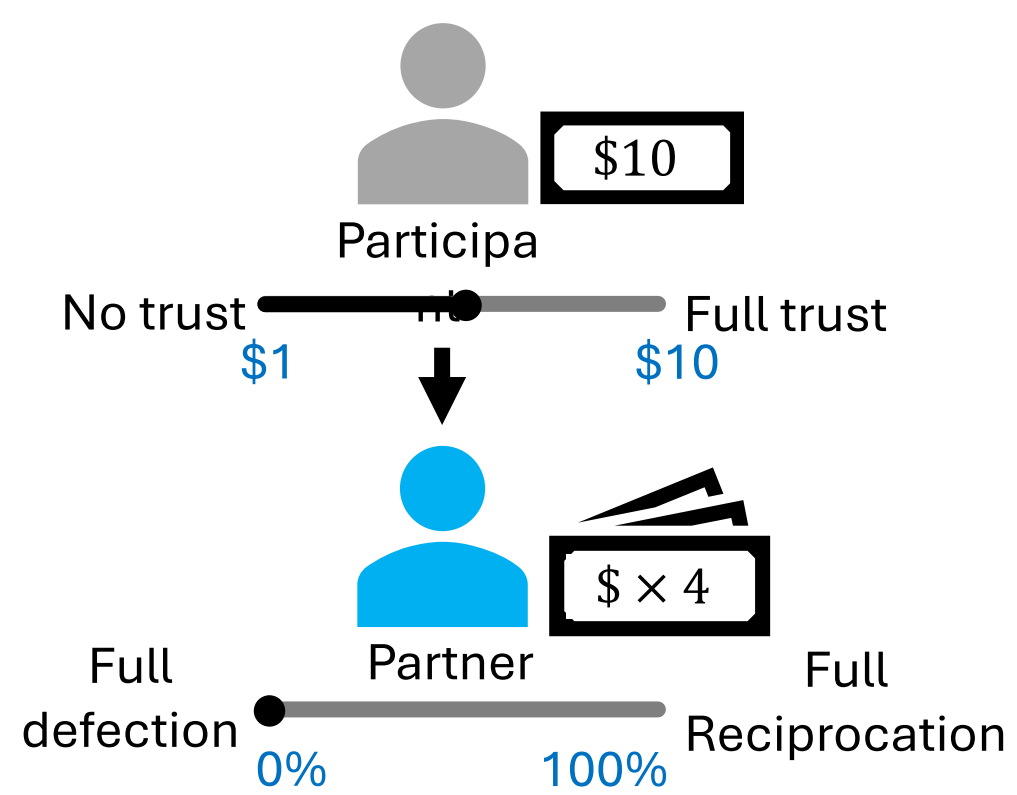
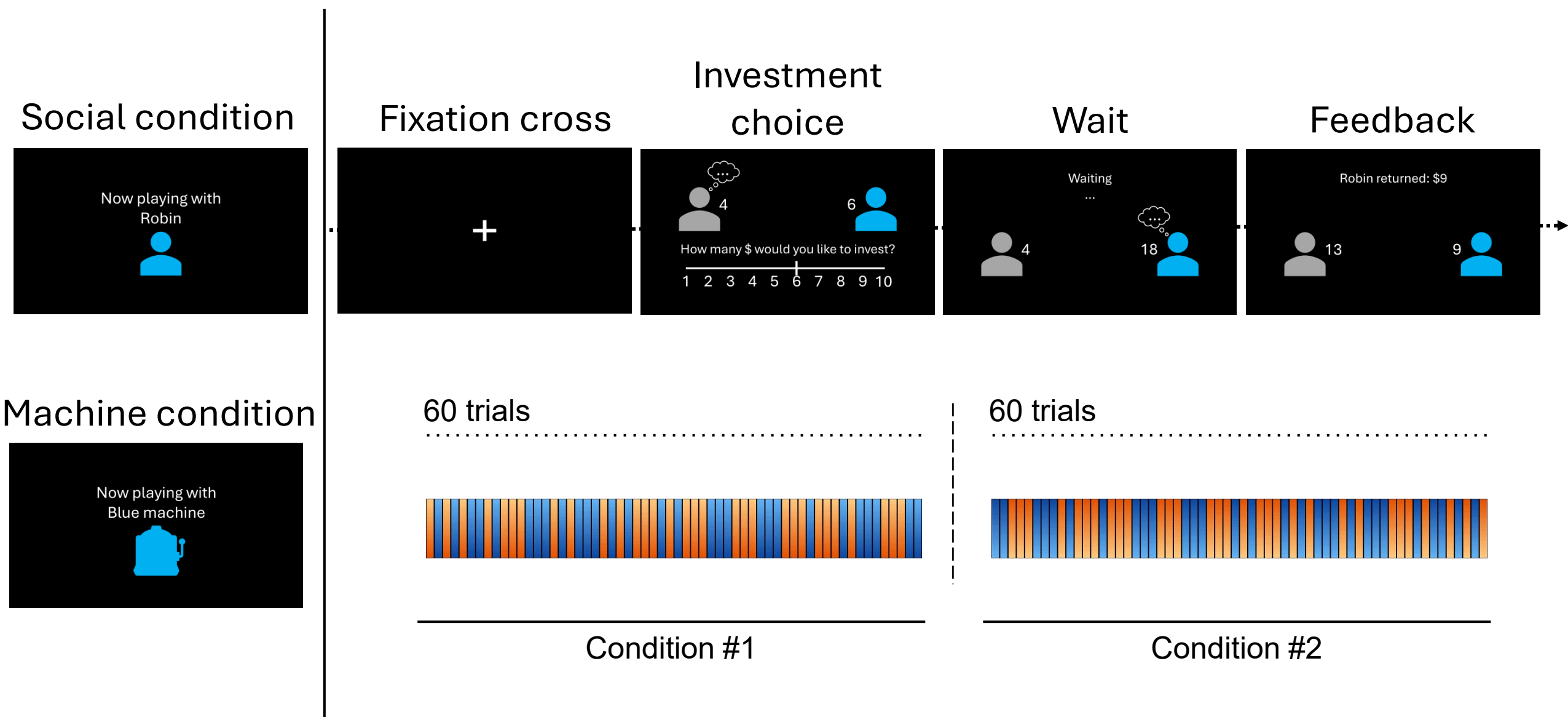
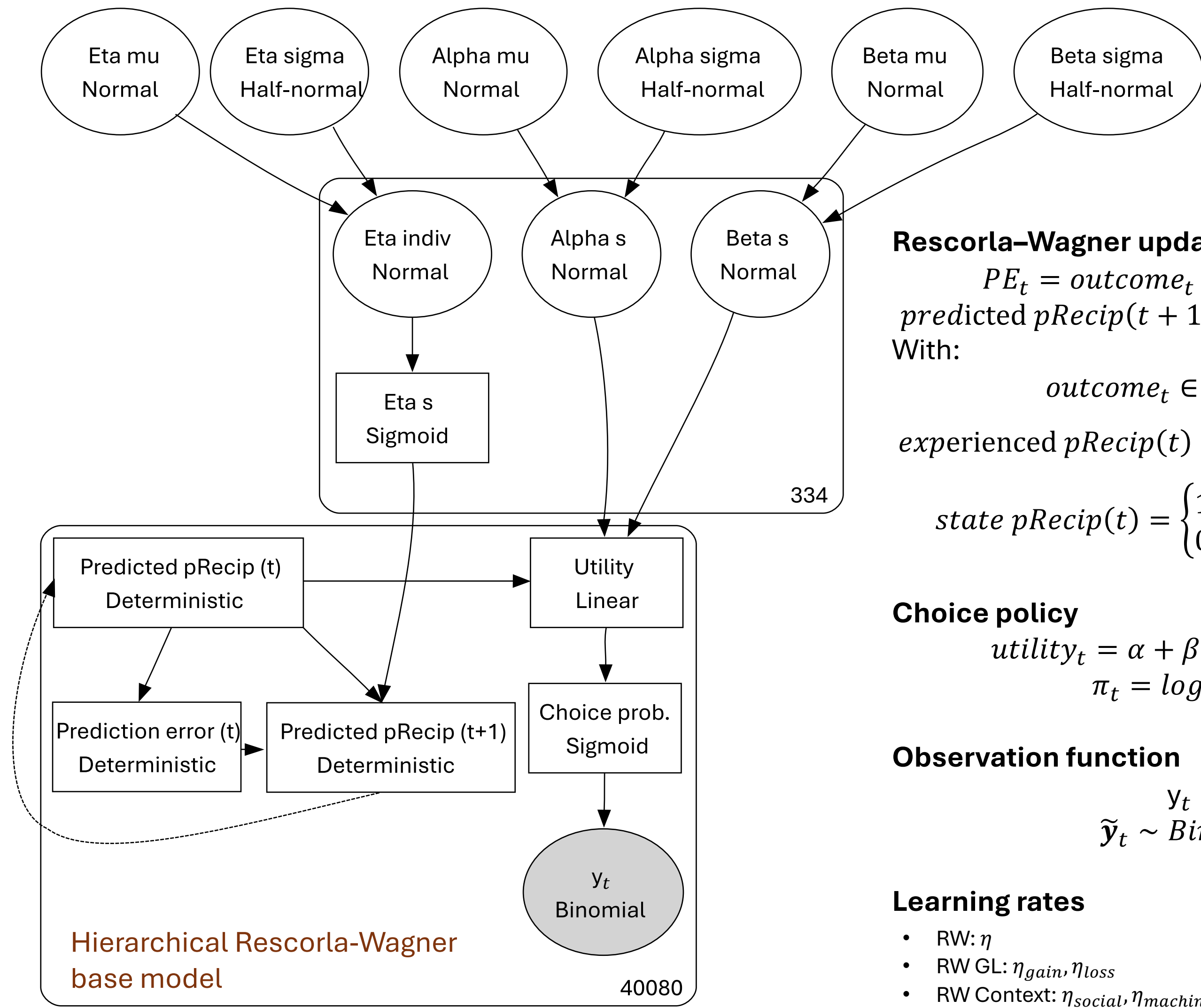
- Trust** is fundamental to social interaction. It is defined as the **willingness to be vulnerable to another being** on the basis of **positive expectations** of their intentions and behaviors (1).

- In repeated interactions, **learning to trust others** involves **cognitive processes** that integrate **uncertainty, context, and potential betrayal**.

- Different strategies** can guide trust behavior. Some rely on **heuristics** (e.g., fixed rules or triggers). Others use **reinforcement learning**, adjusting expected value through associative updates, or **Bayesian belief updating**, which integrates uncertainty into probabilistic inferences about partners' intentions (2, 3).

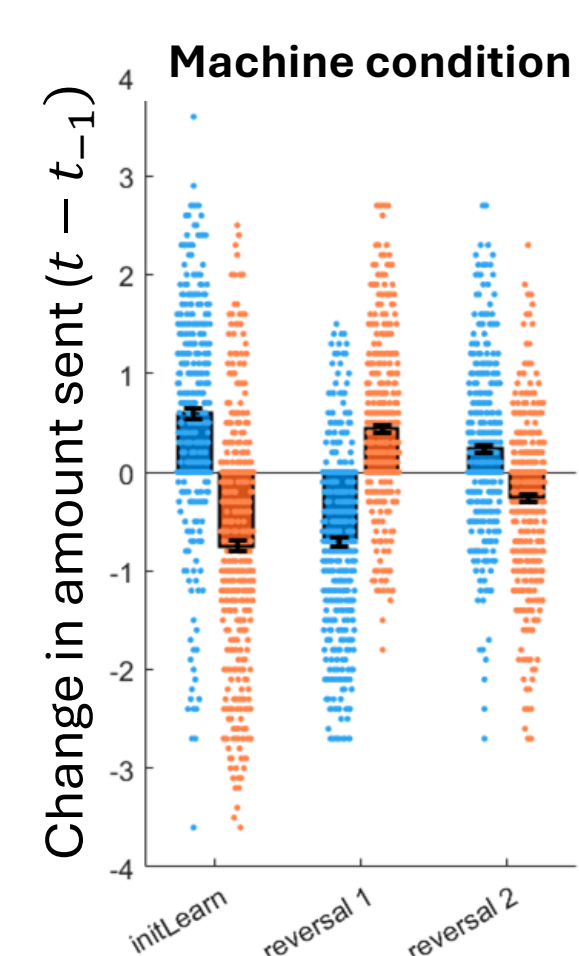
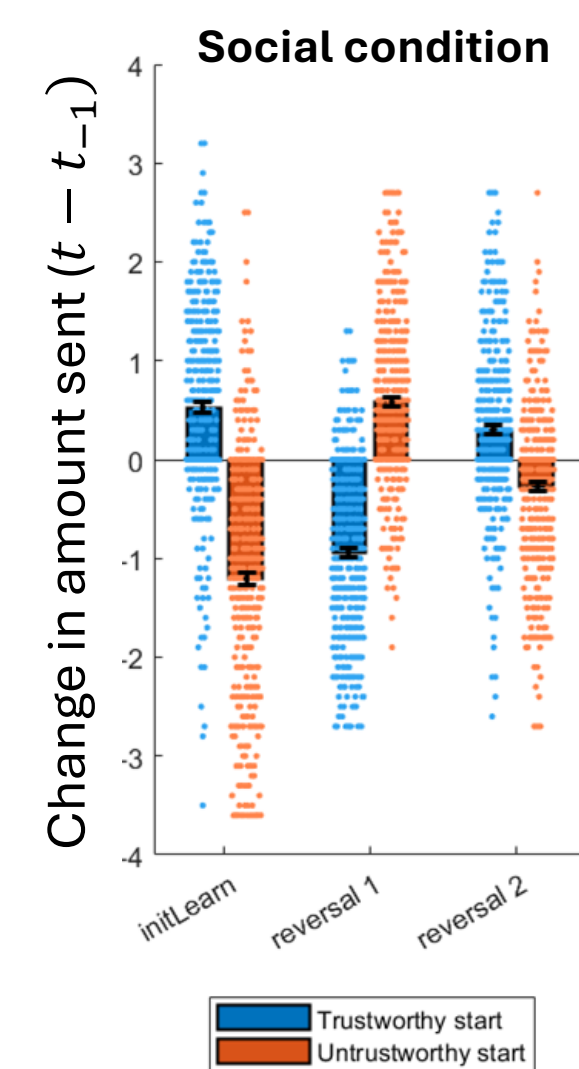
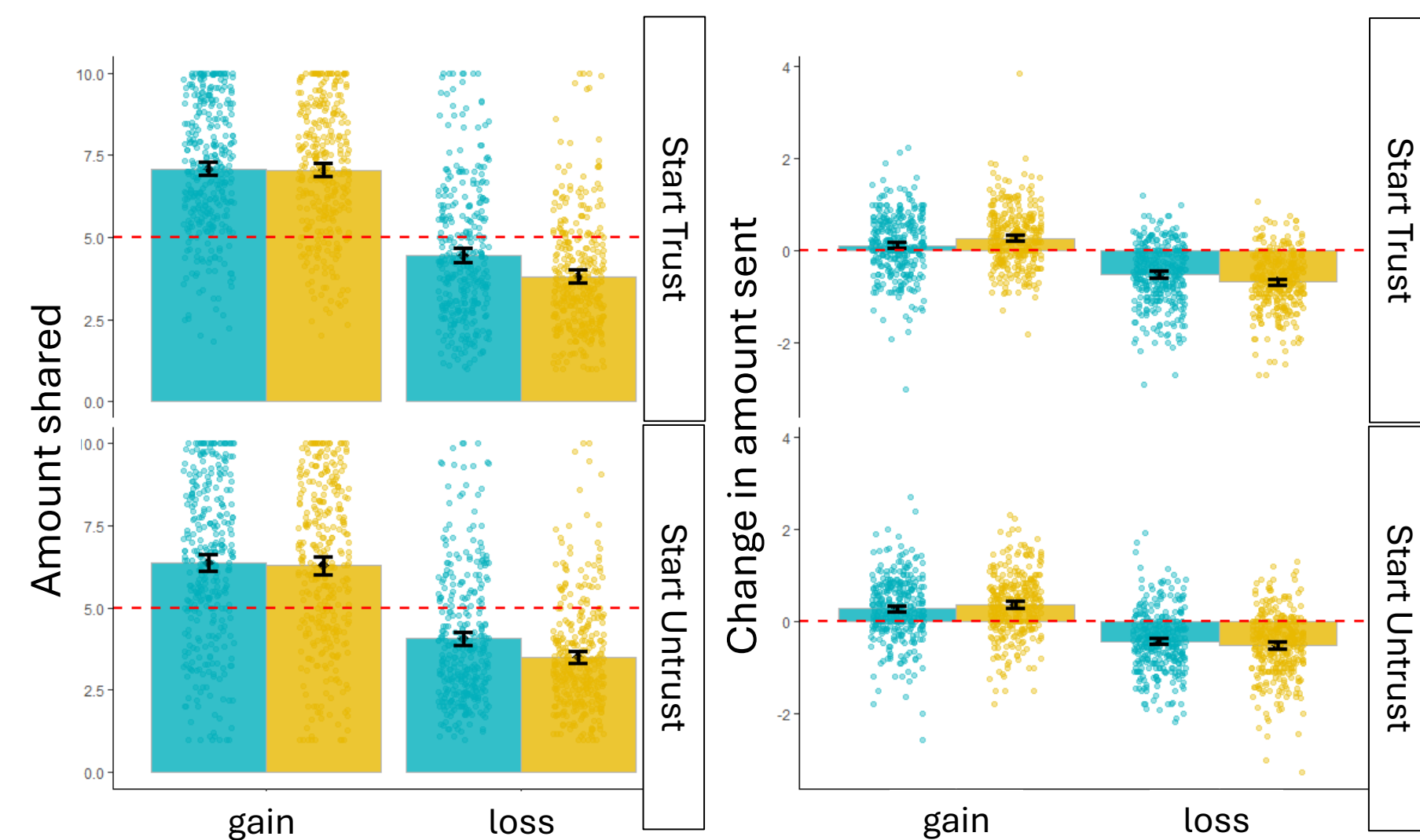
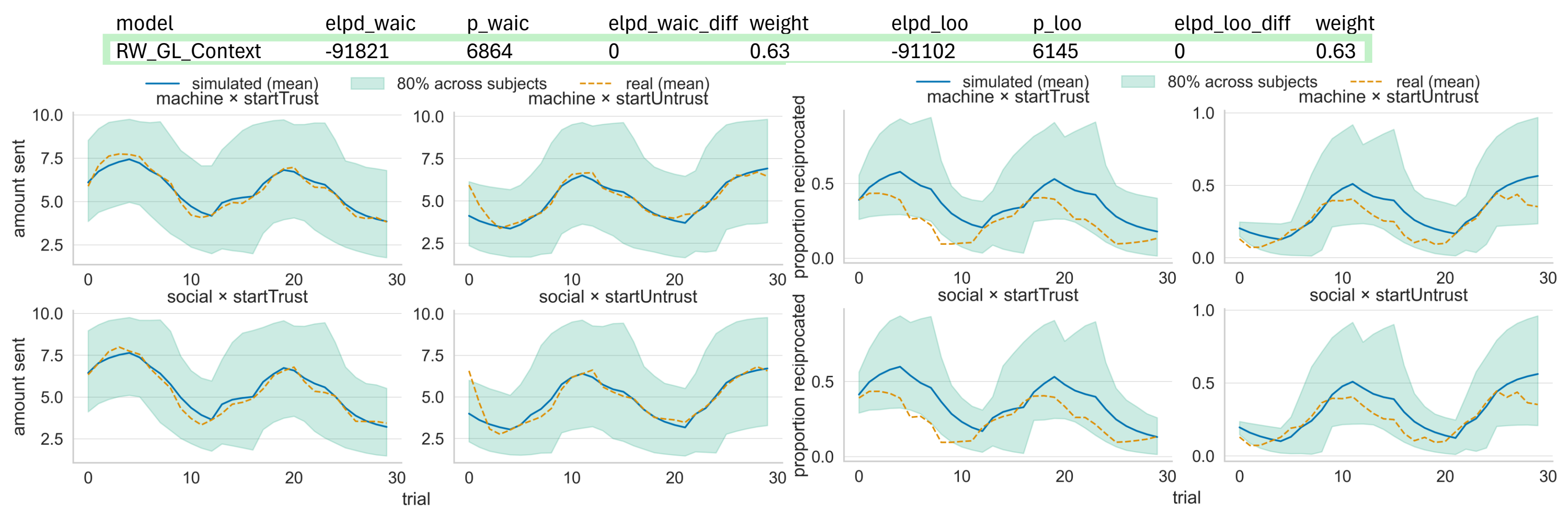
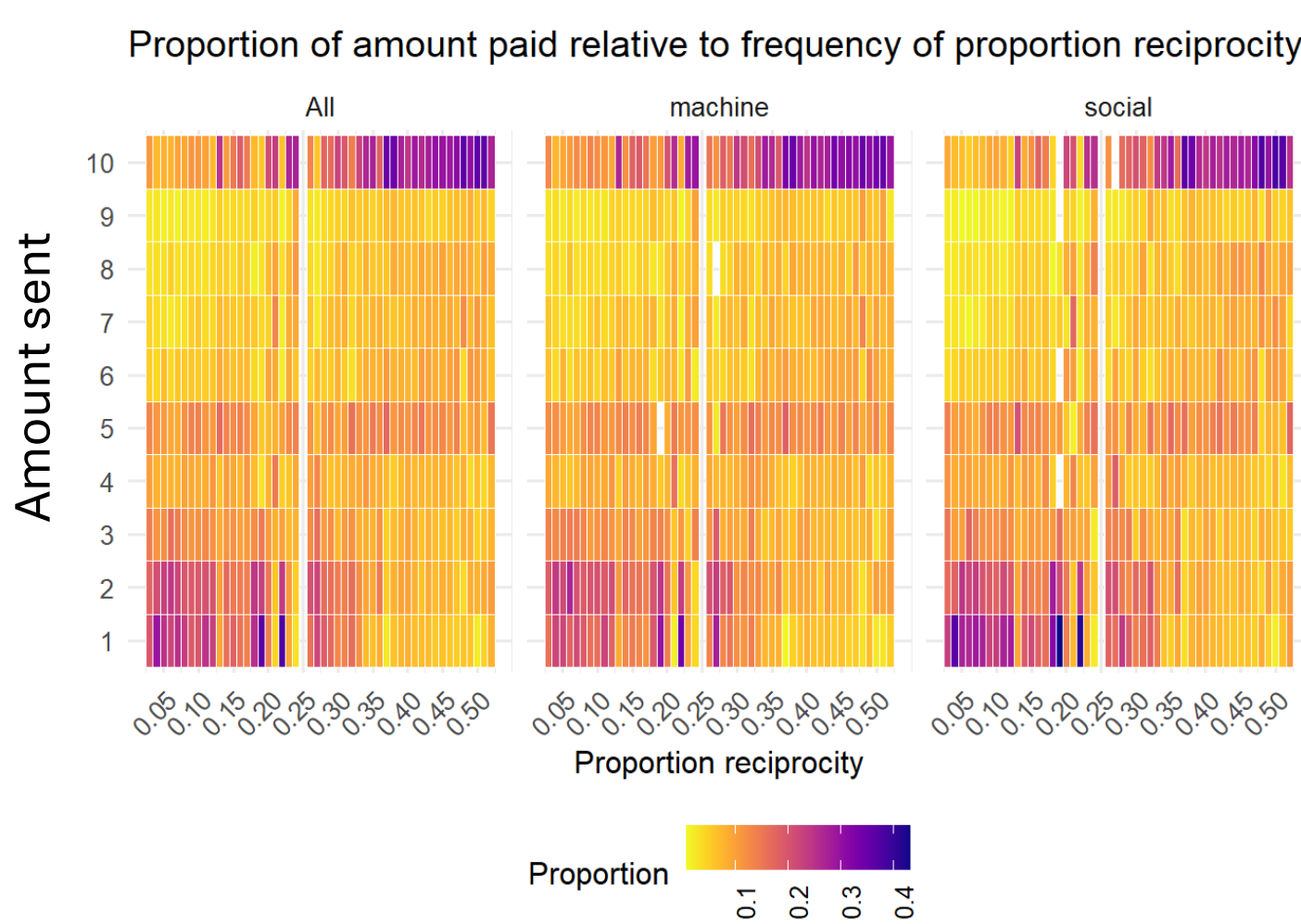
- Individuals differ** in how they deploy cognitive **processes**, which may impact their **strategies**. Yet, this heterogeneity has not been systematically characterized. Here, we aim to do so:

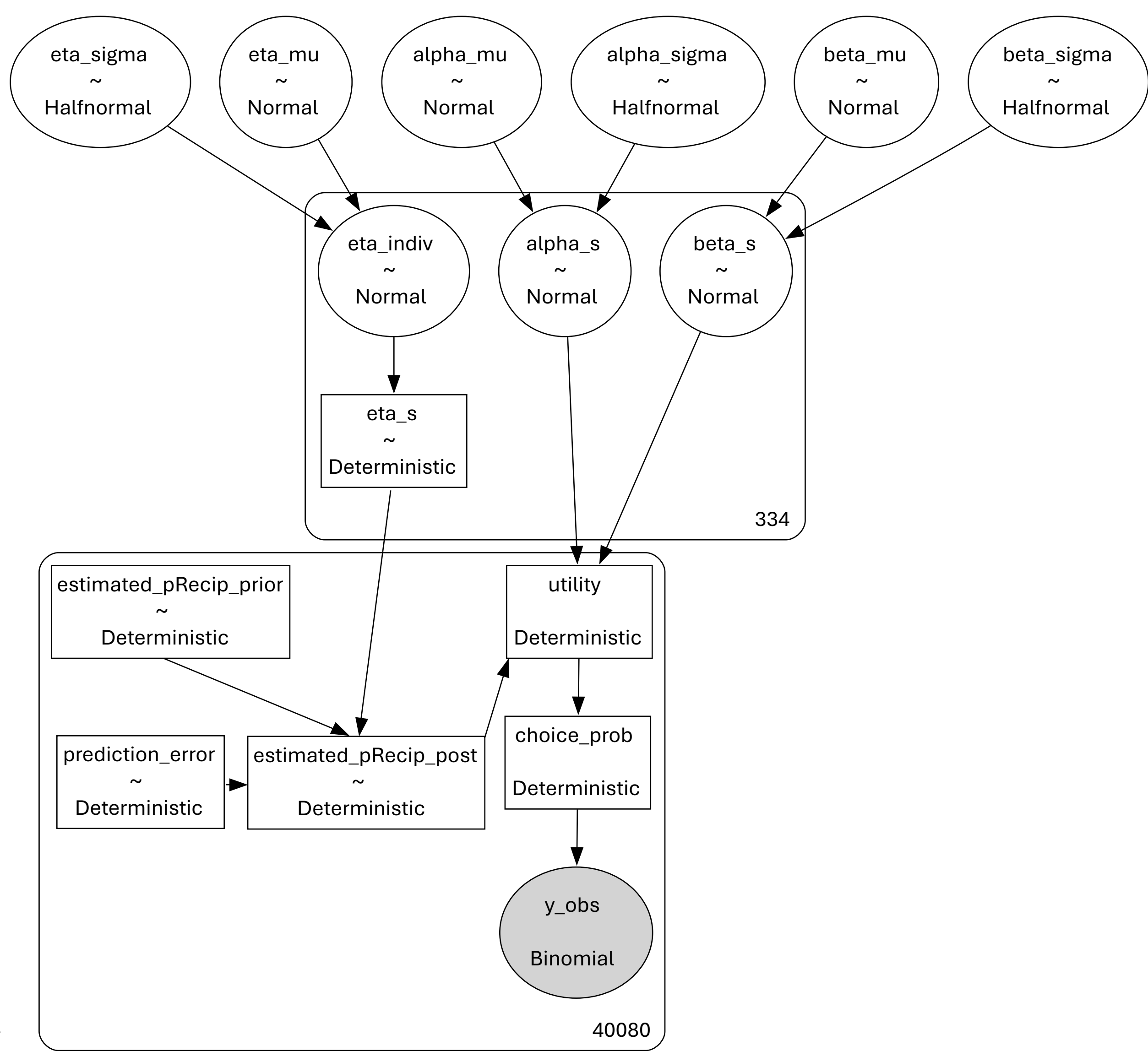
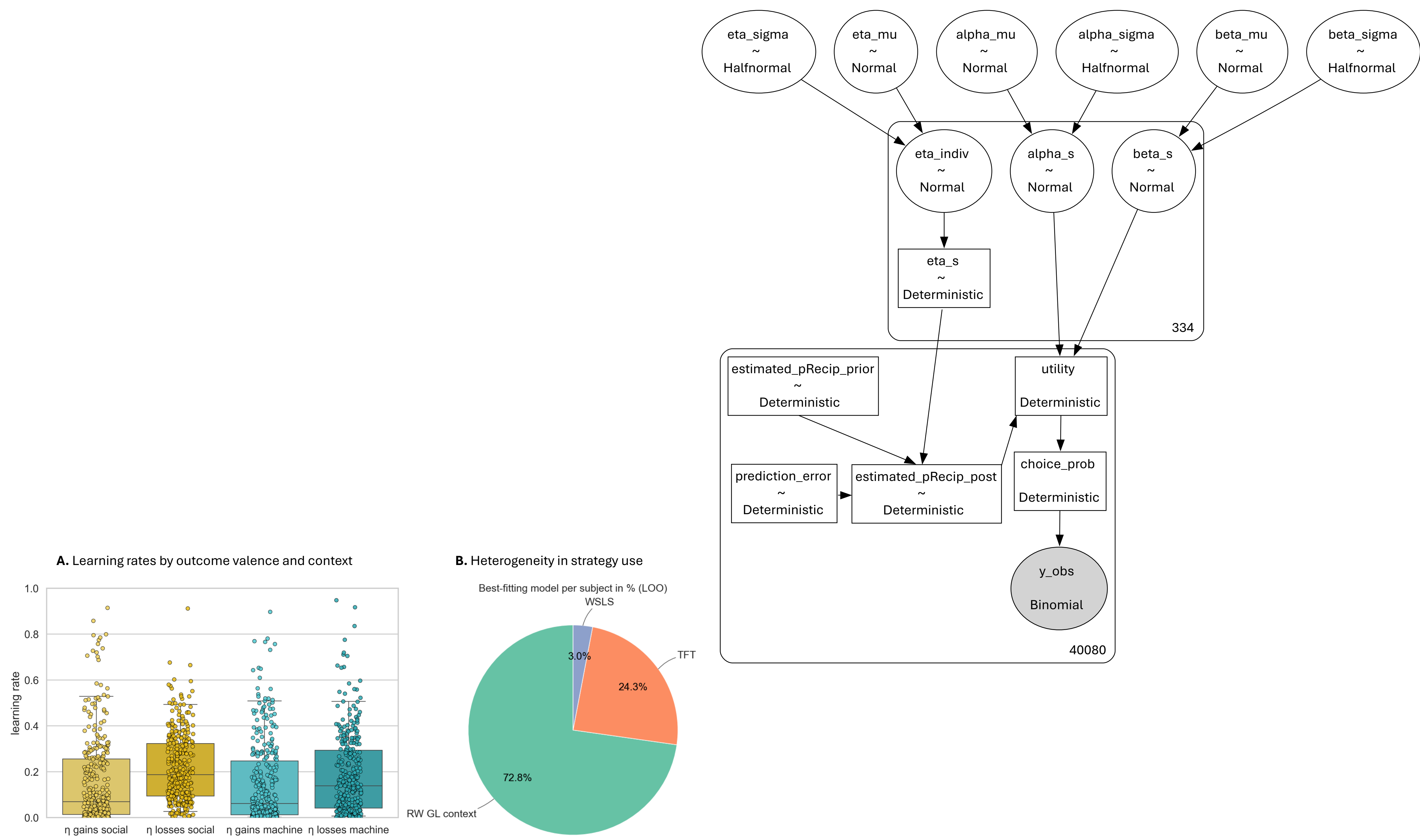
- How do uncertainty, context, and betrayal aversion shape trust across heuristic, reinforcement learning, and Bayesian strategies?
- How do these processes vary across individuals and reveal distinct cognitive profiles?

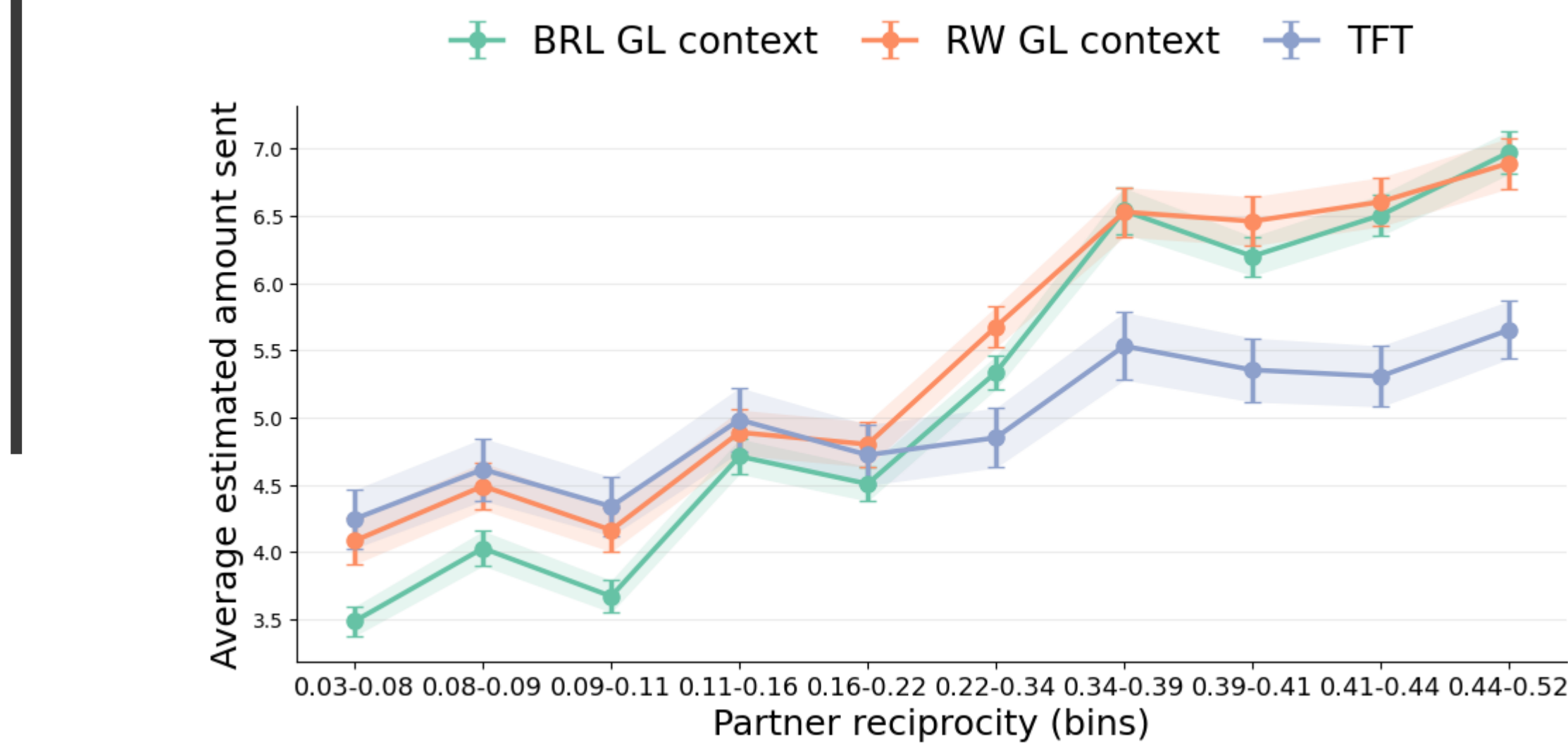


- $N = 334$ ($F=139$, $M=115$), age = 35 ± 11 [19-65] on Prolific
- Indefinitely repeated multinomial Trust game
- 2 pre-programmed partners, 2 pre-prog. machines

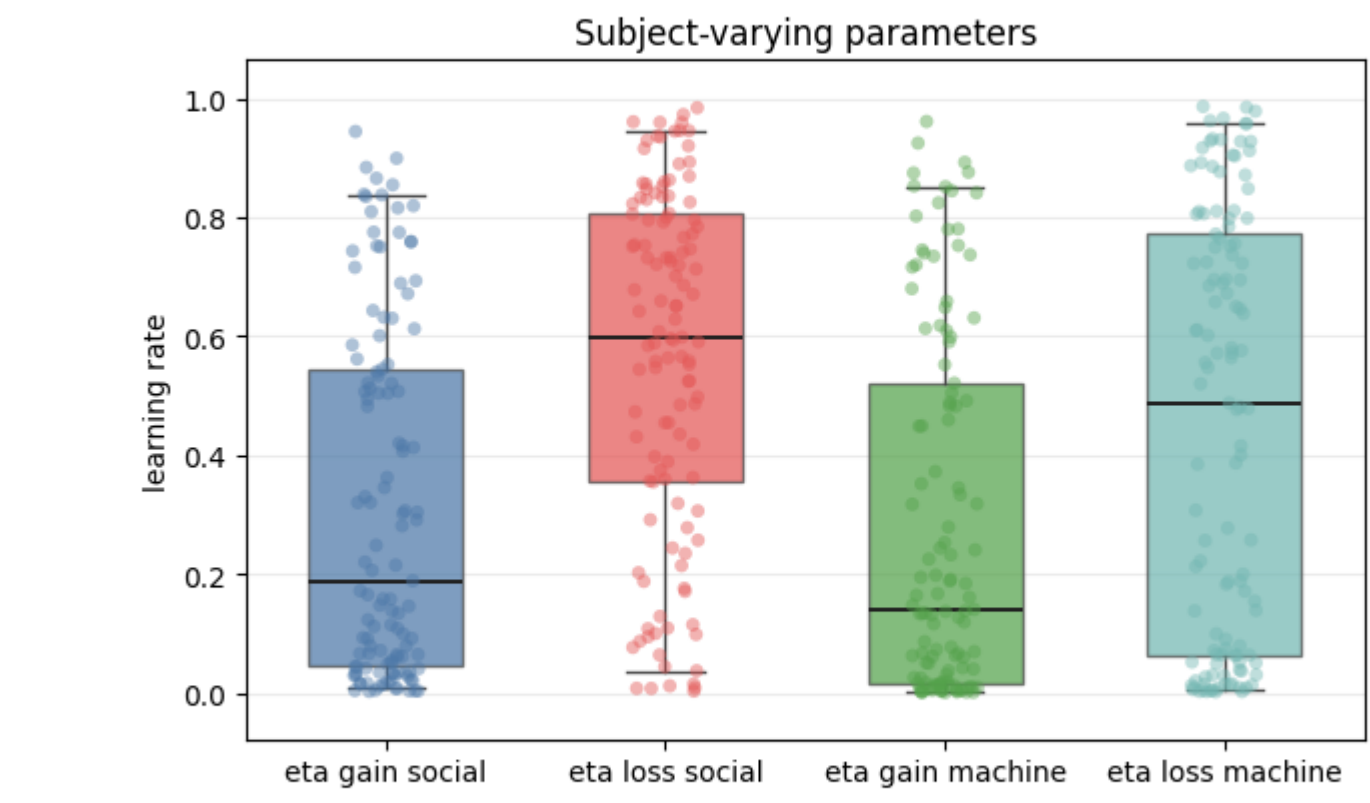
- Subject starts with *Endowment*: $E = 10$
- Subject sends: $s \in \mathbb{N}, 1 \leq s \leq 10$
- Partner receives: $s \times K = 4s$
- Partner reciprocates: $r = 4s \times pr$ where $pr \in [0.02, 0.52]$
- Subject earns: $e = E - s + r$



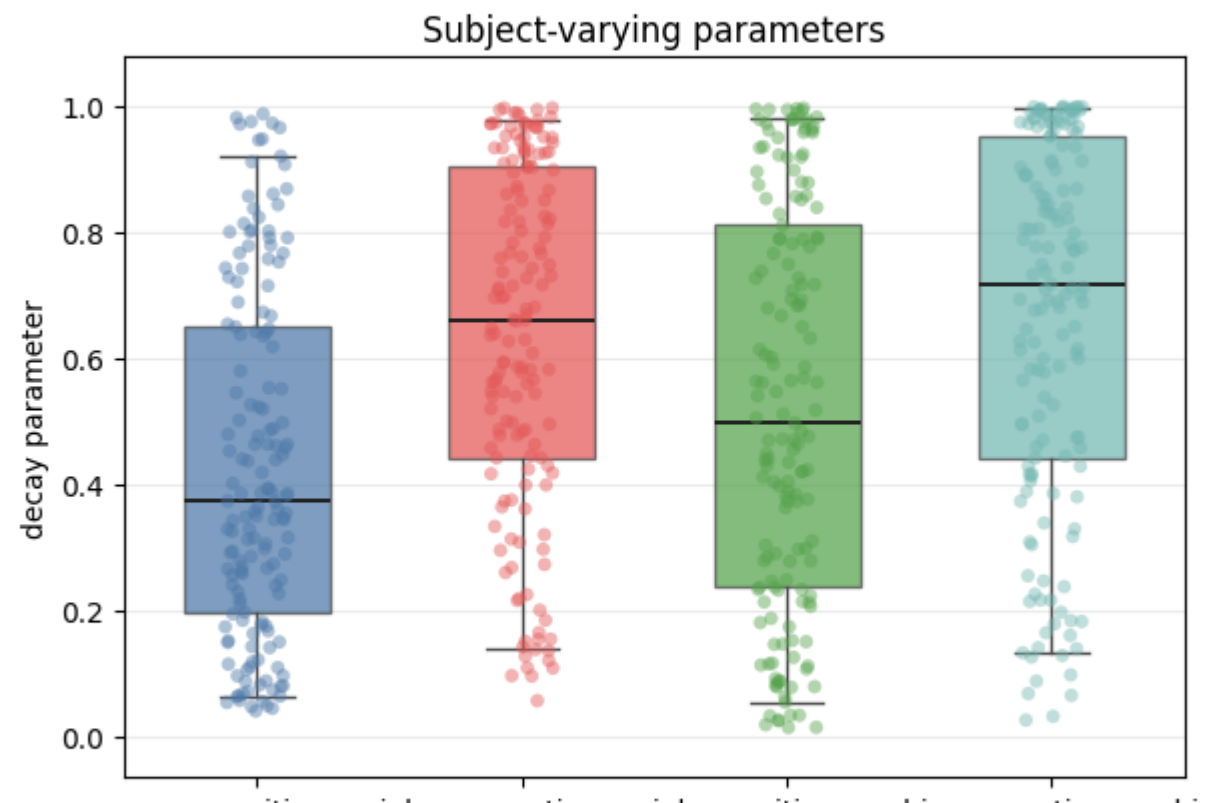




Winner parameter review: Bayesian RL GL Context linear binomial
winner subjects=157 | artifact=idata_brl_gl_context_linear_binomial_blackjax.nc



RL winner review reuses the four learning-rate surfaces within the winner subset.



This panel compares the four BRL gain/loss-context decay surfaces.

